

Order Diptera

Flies

Flies have only a single pair of wings (although some ectoparasitic species are wingless). The hindwings are reduced to small, club-shaped balancing organs called halteres. Flies are ecologically essential as pollinators, parasites, predators, and decomposers, but many damage crops, and the disease-carrying species have a huge impact on wild and domesticated animals, as well as on humans.

There are 130 families and 150,000 species of flies, divided into 2 suborders, Nematocera (26 families) and Brachycera (104 families). The former are delicate, with threadlike antennae and slender bodies, while the latter are more robust, with stout antennae. All have complete metamorphosis: larvae ("maggots") may have a reduced head. Larval feeding habits are among the most diverse in insects.

Family Agromyzidae

Leaf-mining flies

Occurrence 2,500 spp. worldwide; on leaves, stems, seeds, roots of a wide range of plants, including crops

Habitat Terrestrial

This family gets its name from the ability of larvae to chew mines (channels) in leaves. These crop pests also feed inside stems, seeds, or roots. Gray, black, or greenish yellow, they vary from 1/32 to 1/4 in (1–6 mm) in length. Females have a pointed ovipositor, and lay up to 50 eggs a day inside plant tissue. These flies may live for as little as 2 weeks in tropical areas.



HEXOMYZA SP.
These species occur throughout the Northern Hemisphere, and in South Africa and Australia. The larvae make galls on the twigs of poplars.

Family Calliphoridae

Blow flies

Occurrence 1,200 spp. worldwide; on flowers, vegetation, carcasses, cooked and raw food

Bluebottles and greenbottles are familiar members of this widespread and sometimes troublesome family of flies. Like their many relatives, they have stout, bristle-covered bodies, up to 1/2 in (1.5 cm) long, a metallic sheen, particularly on the abdomen, and a noisy, buzzing flight. Blow flies breed in decaying animal matter, dung, or rotten fishes. Adults feed on liquids from rotting material, nectar, or fruit. The larvae of some species are predators of ants, termites, and the larvae and eggs of other insects. Several species burrow into human flesh and many carry diseases that they can transmit to livestock or humans.



POLLENIA RUDIS
Common in the Northern Hemisphere, cluster flies congregate in large numbers in attics and unheated buildings. In the spring, adult females lay their eggs on earthworms or near fresh worm casts.



CALLIPHORA VOMITORIA
The bluebottle fly is common in the countryside in the Northern Hemisphere. Females may lay many hundreds of eggs in their lifetime.

CALLIPHORA VICINA
This widespread species is common in towns and cities, where the maggots develop inside the corpses of dead rats, mice, and pigeons.

Family Asilidae

Robber flies

Occurrence 5,000 spp. worldwide, especially in tropical and subtropical regions; in a variety of microhabitats



This is a family of predatory flies; both the adults and larvae are predatory, although the latter eat decaying matter as well. Robber flies use their forward-pointed, stout and sharp proboscis to stab prey and inject a paralyzing saliva. They then suck up the body contents of the paralyzed insect. They are 1/8–2 in (0.3–5 cm) long, either slender or stout and beelike. Their head is slightly hollow between the eyes, and the face has a tuft of long hairs. Eggs are laid in soil, rotting wood, or inside plants.



BLEPHAROTES SPLENDIDISSIMUS
This large Australian robber fly has platelike tufts of hair at the sides of its flat abdomen.

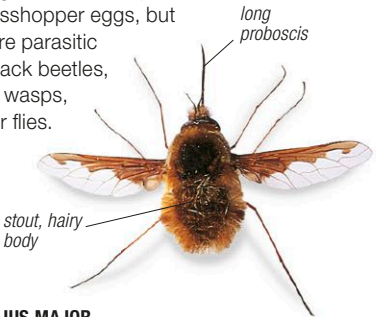
Family Bombyliidae

Bee flies

Occurrence 5,000 spp. worldwide, in open, tropical, or subtropical semiarid regions; around flowers, on ground



Bee flies are stout and hairy insects that strongly resemble bumble bees (see p.573). Between 1/16–1 1/4 in (0.2 and 3 cm) long, they are brown, red, or yellow, sometimes with bright markings. They have narrow wings, and when flying produce a high-pitched whine. Like bees, they have a long proboscis that they use to suck nectar from flowers. However, instead of landing on flowers as bees do, bee flies hover in front of them, steadying themselves against the petals with their front legs. Some bee fly larvae eat grasshopper eggs, but most are parasitic and attack beetles, moths, wasps, or other flies.



BOMBYLIUS MAJOR
The large bee fly is common and inhabits meadows, hedgerows, and open areas from Europe to China, and North Africa.

Family Cecidomyiidae

Gall midges

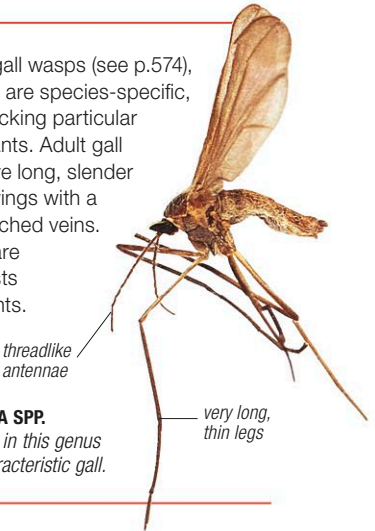
Occurrence 5,000 spp. worldwide; anywhere near decaying matter, fungi, a wide range of plants



With bodies typically less than 5/32 in (4 mm), gall midges are seldom seen, but their larval homes—plant galls—are much easier to spot. Looking like miniature fruit or malformed buds, the galls are growths triggered by some gall midges, for protection and as a source of

food. Like gall wasps (see p.574), gall midges are species-specific, always attacking particular kinds of plants. Adult gall midges have long, slender legs, and wings with a few unbranched veins. A number are serious pests of crop plants.

ASPHONDYLIA SPP.
Each species in this genus makes a characteristic gall.



Family Ceratopogonidae

Biting midges

Occurrence 4,000 spp. worldwide, mainly in the Northern Hemisphere; near margins of water bodies

Habitat All

Also called punkies or no-see-ums, these small flies—under 3/16 in (5 mm) long—have a bite that causes severe irritation. They have short, strong legs, dark patterns on their wings, and piercing mouthparts shaped to suck blood and insect body fluids. They mate while flying in a swarm. Eggs are laid in groups of 30–450, and are covered with a protective, jellylike substance.

Some biting midges transmit worm parasites to humans or carry animal diseases. They are pests but are also important pollinators of tropical crops such as cocoa or rubber.



CULICOIDES IMPUNCTATUS
This notorious European species breeds in boggy areas. The bite of the female common midge is extremely itchy and painful.